

Pset 10

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Problem #11.1:

$$v(t) = 160 \cos(50t) \text{ V and } i(t) = -33 \sin(50t - 30^\circ) \text{ A}$$

$$\text{Convert } i(t) \Rightarrow i_c(t) = 33 \cos(50t + 60^\circ) \text{ A}$$

$$\begin{aligned} p(t) &= v(t) \cdot i(t) = (160)(33) \cos(50t + 60^\circ) \cdot \cos(50t) \\ &= 5280 \cos(50t) \cos(50t + 60^\circ) \\ &= 2640 (\cos(-60^\circ) + \cos(100t + 60^\circ)) \end{aligned}$$

$$p(t) = 1320 + 2640 \cos(100t + 60^\circ) \text{ W.}$$

$$\begin{aligned} P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2} (160)(33) \cos(0^\circ - 60^\circ) \\ &= 1320 \text{ W} \end{aligned}$$

Problem #11.8

KCL at N2:

$$0.5 I_o + I_o = \frac{V_2}{40} \Rightarrow V_2 = 60 I_o$$

$$V_c = V_1 - V_2 = (-j20) I_o$$

$$V_1 = 60 I_o - j20 I_o$$

$$= (60 - j20) I_o$$

KCL at N1:

$$G = \frac{V_1}{j10} + I_o = \frac{60 - j20}{j10} I_o + I_o$$

$$\begin{aligned} \frac{60 - j20}{j10} &= \frac{-j(60 - j20)}{10} = \frac{-20 - j60}{10} \\ &= -2 - j6 \end{aligned}$$

$$\therefore G = (-2 - j6 + 1) I_o = (-1 - j6) I_o$$

$$\Rightarrow I_o = \frac{G}{-1 - j6}$$

$$|I_o| = \frac{G}{\sqrt{1 + 36}} = 0.9864 \text{ A}$$

$$\begin{aligned} I_{40} &= \frac{V_2}{40} \\ &= 1.5 I_o \end{aligned}$$

$$\begin{aligned} |I_{40}| &= 1.5 (0.9864) \\ &= 1.4796 \text{ A} \end{aligned}$$

$$\begin{aligned} \therefore P &= \frac{1}{2} |I_{40}|^2 R \\ &= \frac{1}{2} (1.4796)^2 (40) \\ &= 43.79 \text{ W} \end{aligned}$$

Problem 11.19

Z_{Th} :

$$\begin{aligned} G || (3 + j1) &= \frac{G(3 + j1)}{9 + j1} = \frac{18 + j6}{9 + j1} \\ &= 2.0486 + j0.4390 \Omega \end{aligned}$$

$$\begin{aligned} R &= |Z_{Th}| \\ &= \sqrt{(2.0488)^2 + (1.5610)^2} \\ &= 2.576 \Omega \end{aligned}$$

$$\begin{aligned}\therefore Z_{Th} &= 2.0488 + j0.4390 - j2 \\ &= 2.0488 - j1.5610 \Omega\end{aligned}$$

$$V_{Th} = V_A:$$

$$\begin{aligned}V_{Th} &= 33(2.0488 + j0.4390) \\ &= 69.14 \angle 12.1^\circ \text{ V}\end{aligned}$$

$$\begin{aligned}|Z_{Th} + R|^2 &= (2.0488 + 2.576)^2 + (1.5610)^2 \\ &= 23.823\end{aligned}$$

$$\therefore |I|^2 = \frac{|V_{Th}|^2}{|Z_{Th} + R|^2} = \frac{(69.14)^2}{23.823} = 200.7$$

$$\begin{aligned}\therefore P_{max} &= \frac{1}{2} |I|^2 R = \frac{1}{2} (200.7) (2.576) \\ &= 258.5 \text{ W}\end{aligned}$$

Problem #11.22

$$\begin{aligned}I_{rms} &= \sqrt{I_{dc}^2 + \frac{I_m^2}{2}} \\ &= \sqrt{2^2 + \frac{2^2}{2}} \\ &= \sqrt{6} \text{ A}\end{aligned}$$

Problem #12.1:

$$V_p = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 231 \text{ V}$$

a)

$$V_{an} = 231 \angle -30^\circ \text{ V}$$

$$V_{bn} =$$